

Data Profiler - Theory Notes

1. Data Profiling

Data profiling means examining a dataset to understand its structure and quality. It can reveal missing values, duplicates, data types, statistics and relationships.

2. DataFrame

A DataFrame is a table-like structure in pandas made up of rows and columns.

3. Tensor

A tensor is a multi-dimensional array of numbers. Scalars, vectors and matrices are simple cases of tensors.

4. Data Acquisition

Data acquisition means collecting data from sources such as CSV, JSON, SQL databases and APIs.

5. Data Cleaning

Data cleaning prepares raw data for analysis by handling missing values, duplicate records, wrong types and unnecessary columns.

6. Missing Values

A missing value means that information is not available. Numerical missing values can sometimes be replaced by a mean or median, while categorical values can use a mode.

7. Exploratory Data Analysis

EDA means exploring data with statistics and graphs. Univariate uses one variable, bivariate uses two, and multivariate uses more than two.

8. Correlation

Correlation describes the strength and direction of a linear relationship between numerical variables. It does not prove causation.

9. Machine Learning Problem

Customer Purchase Prediction can be treated as a regression problem because Purchases is numerical.

10. Duplicate Records

Duplicate records are repeated rows. They can be checked with `duplicated()` and removed with `drop_duplicates()`.